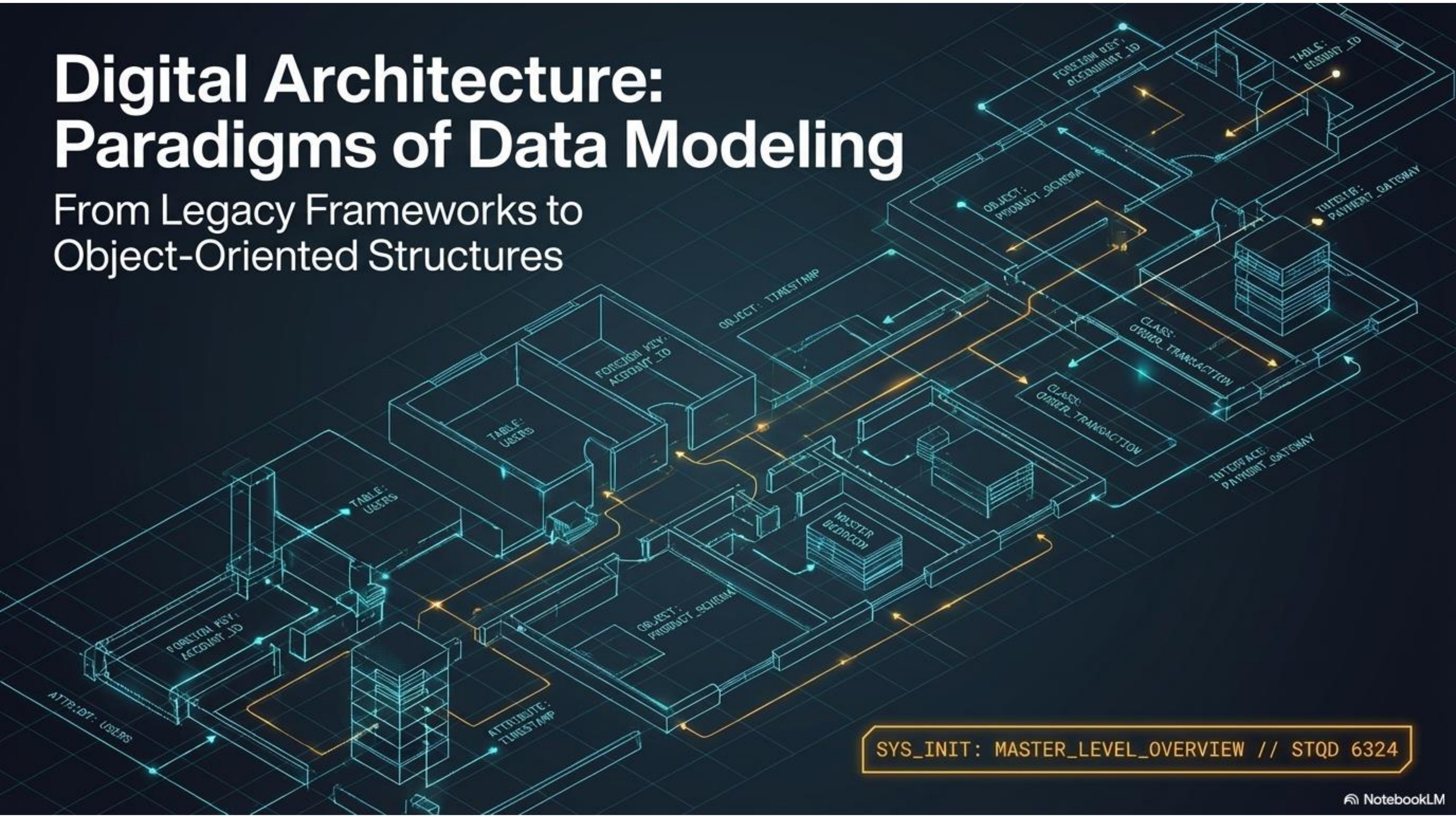
Digital Architecture: Paradigms of Data Modeling

From Legacy Frameworks to Object-Oriented Structures



SYS_INIT: MASTER_LEVEL_OVERVIEW // STQD 6324

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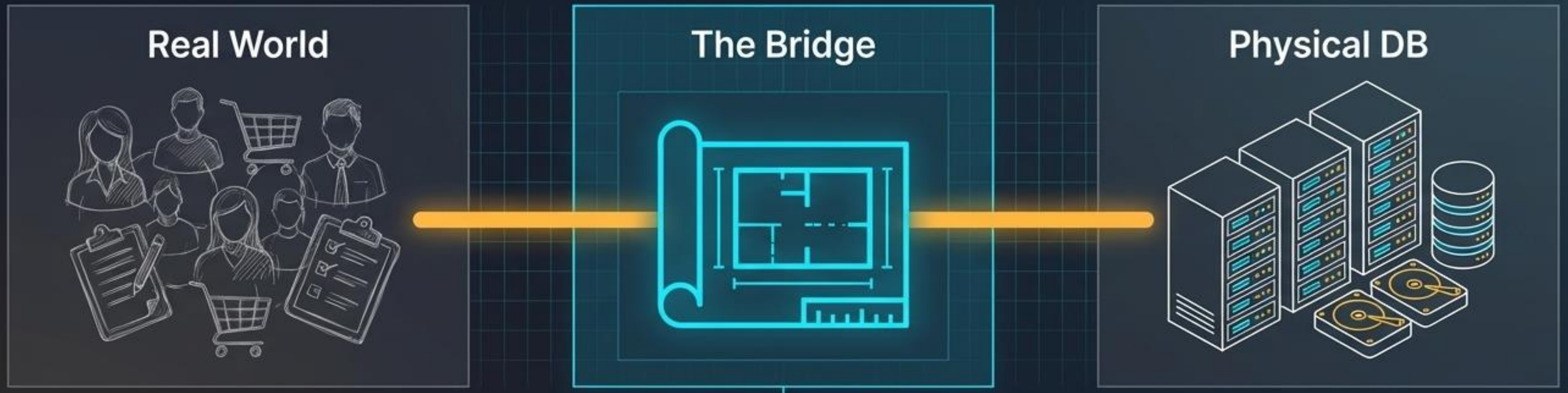
The Conceptual Bridge

A data model acts as the architectural blueprint between real-world events and physical hardware storage. It represents data exclusively as the user sees it, rendering it entirely independent of underlying hardware or software constraints.



01. ERD (Entity-Relationship Diagram)

The visual communication tool for stakeholders.

02. Data Dictionary

The comprehensive metadata document (detailing objects, types, rules) required by developers to construct the physical database.

Column	Data Type	Description
employee_id	int	Primary key of a table
first_name	nvarchar(50)	Employee first name
last_name	nvarchar(50)	Employee last name
nin	nvarchar(15)	National Identification Number
position	nvarchar(50)	Current position title, e.g. Secretary
dept_id	Int	Employee department. Ref. Departments

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The Architect's Toolkit

Entity

A distinguishable, unique object in the real world used to collect and store data.

`TABLE: CUSTOMER_RECORD`

Attribute

A characteristic or descriptive property of a specific entity.

`VARCHAR: last_name`

Relationship

The structural association among entities. Expressed strictly as 1:1, 1:M, or M:N.

`CONNECTIVITY: 1..*`

Constraint

A strict rule ensuring logical data integrity across the model.

`CHECK (GPA BETWEEN 0.00 AND 4.00)`

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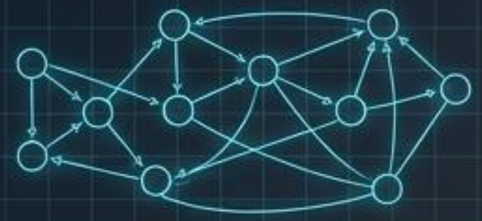
Legacy Constraints: The Pre-Relational Era

Hierarchical Model	 **Network Model**
Structure: Ordered Tree (Top-Down)	Structure: Complex Graph
Navigation: “Left-list” path traversal; relies on physical parent-to-child proximity to minimize disk I/O time.	Navigation: Utilizes “Sets” containing at least two record types (Owner and Member).
Capacity: Strictly limited to 1:M relationships. Uses “Virtual Records” (logical pointers) to mitigate extreme data replication.	Capacity: Handles complex M:N relationships; a member can have multiple owners.
Fatal Flaw: Deletion anomalies (deleting a parent automatically cascades to child) and complete lack of structural independence.	Fatal Flaw: Overwhelming system complexity and an inflexible, pointer-based architecture that makes reorganization exceptionally difficult.

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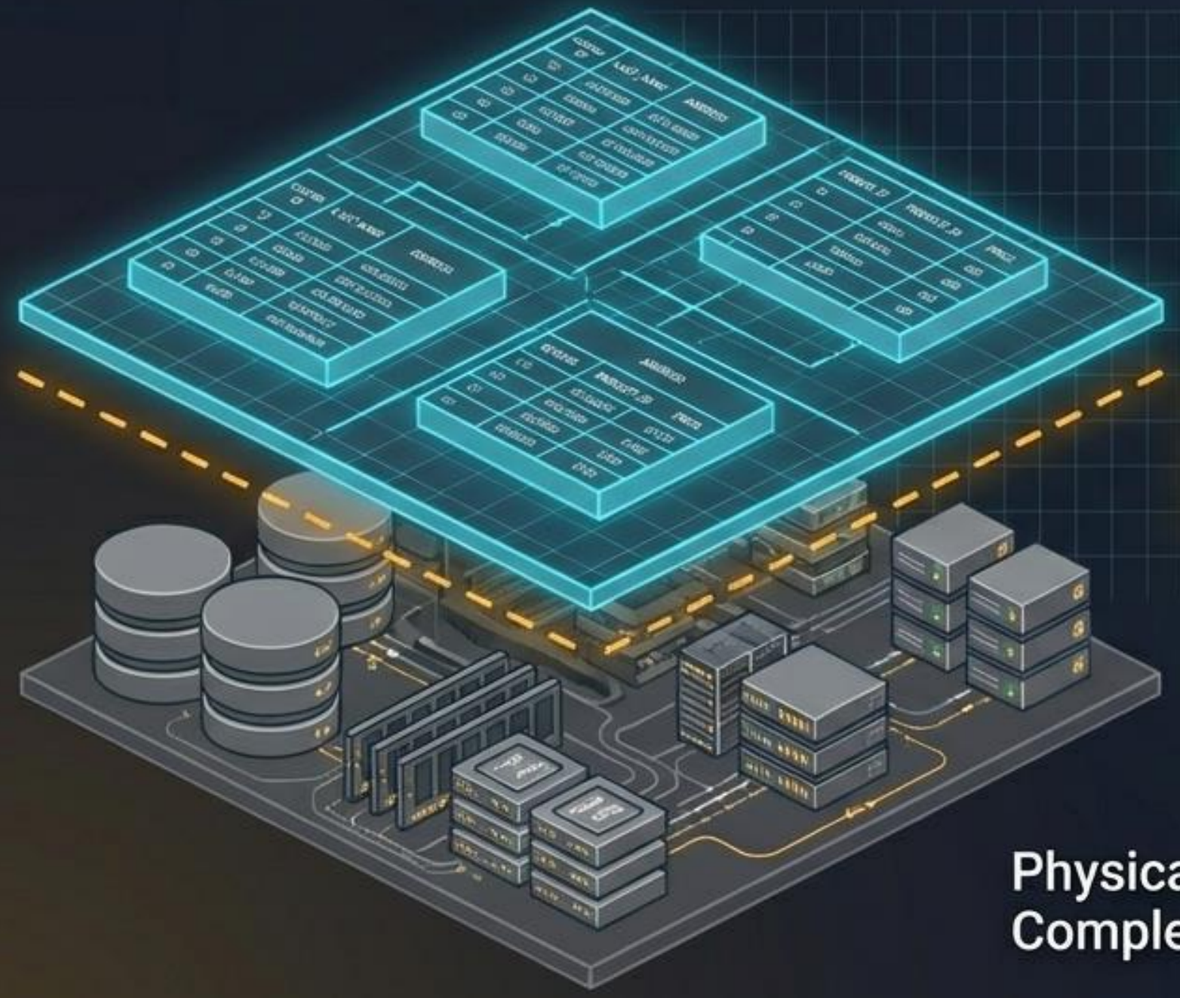
The Relational Paradigm Shift (1970)

Developed by E.F. Codd, the relational model initiated a database revolution by achieving true Structural Independence.



Core Mechanic

The Relational Database Management System (RDBMS) entirely absorbs the complex physical details of storage.

The Result

Designers and users operate exclusively in a human-logical environment, interacting with data conceptually rather than mechanically via Ad Hoc Queries (SQL).

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Overcoming Islands of Information

AGENT

AGENT_CODE	AGENT_LNAME	AGENT_FNAME	AGENT_INITIAL	AGENT_AREACODE	AGENT_PHONE
S01	Anneson	Lwon	A	317	990-339-3800
S02	Manity	John	B	320	958-803-0300
S01	Betward	Maria	N	317	996-864-8370
S04	Janrson	Canry	A	330	981-437-3750
S01	Jomirman	Davia	A	332	956-334-3750

Link through AGENT_CODE

CUSTOMER

CUS_CODE	CUS_LNAME	CUS_FNAME	CUS_INITIAL	CUS_AREACODE	CUS_PHONE	CUS_RENEW_DATE	AGENT_CODE
S01	Sursson	Kum	A	372	899-432-3880	12/25/2021	S01
S02	Buntamd	Konmy	H	329	993-324-0320	12/21/2021	S02
S03	Monson	Bernon	F	290	948-373-3896	09/24/2021	S02

Concept:

Relational schemas link independent data matrices (tables) by sharing common entity characteristics (Keys).

The Problem Addressed:

Prevents isolated islands of data (e.g., Hospital Billing vs. Hospital Personnel systems) by weaving them into a unified, query-able architecture.

WARNING: Substantial hardware and system software overhead required compared to legacy pointer systems.

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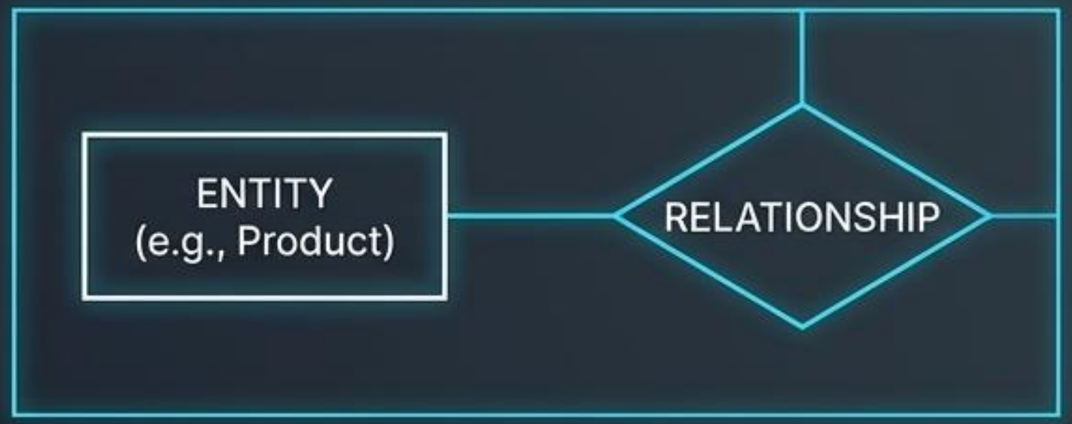
The Drafting Language: Entity-Relationship Models

Introduced by Peter Chen in 1976, the ER Model provided a widely accepted graphical tool for data modeling, perfectly integrated with the relational paradigm.

The Data

ENTITY_ID	ENTITY_NAME	DESCRIPTION
E001	Product	Inventory item
E002	Order	Customer order
E003	Customer	User profile

The Draft



Entity → Maps to a full Relational Table.

Entity Instance (Occurrence) → Maps to a single Row.

Connectivity → Diamond or line endpoints denoting relationship types.

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The Notation Rosetta Stone: Chen vs. Crow's Foot

Chen Notation

[1:M]: PAINTER paints PAINTING

[M:N]: EMPLOYEE learns SKILL

[1:1]: EMPLOYEE manages STORE

Crow's Foot Notation

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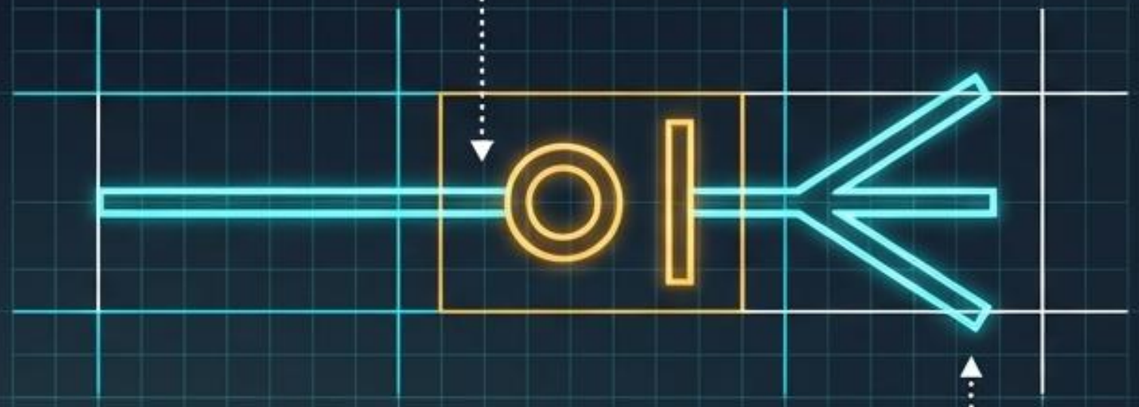
Deciphering Crow's Foot Syntax

Cardinality constraints are embedded directly into the line endpoints, reading from the inside out.

Inner Symbol (Minimums)

Ring (O) = Minimum zero (Optional).
Dash (|) = Minimum one (Mandatory).



Outer Symbol (Maximums)

Dash (|) = Maximum one.
Foot (<) = Maximum many (infinite).

	Minimum zero, maximum one (optional)
	Minimum one, maximum one (mandatory)
	Minimum zero, maximum many (optional)
	Minimum one, maximum many (mandatory)

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Next-Generation Architecture: The Object-Oriented Shift

Based on the Semantic Data Model (SDM, 1981), OODM evolved to handle complex data types that traditional RDBMS struggled to manage.

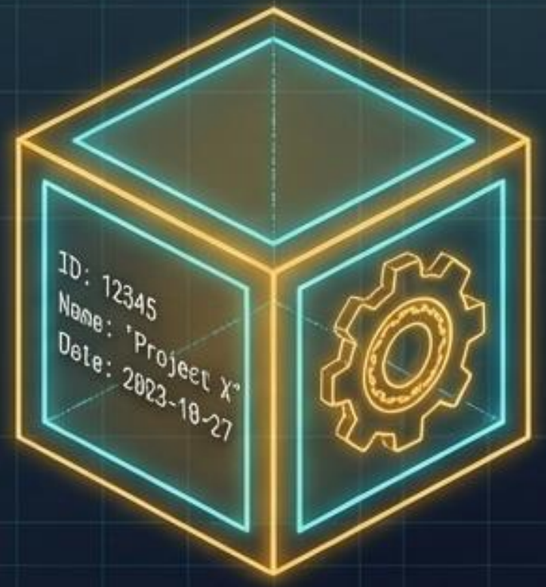
Relational Entity

```
ID: 12345
Name: "Project X"
Date: 2023-10-27
```

Describes factual content and data only.

Methods/Operations

OO Object



An autonomous building block containing factual content, relationships, AND operations (behaviors) in a single encapsulated structure.

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Vertical Expansions: Classes & Inheritance

OO Data Model

ER Model

The Mechanics:

Classes: Objects sharing similar characteristics are grouped into modular classes.
Inheritance: The architectural ability of an object to inherit attributes and methods from classes above it in the hierarchy.

The ROI:

As transaction requirements grow sophisticated, OODM guarantees high data integrity, semantic richness, and crucial code reusability.

ERROR_TRACE: High system overhead | Steep learning curve | Complex navigational access

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Master Architectural Synthesis

	Relational	Entity-Relationship	Object-Oriented
Primary Building Block	The Matrix Table (Rows/Columns)	The Conceptual Entity	The Encapsulated Object
Data & Behavior Integration	Strictly Separated (Logic vs. Storage)	Data only (Visual Draft)	Fully Integrated (Data + Methods)
Primary Strength	Structural Independence & Ad Hoc SQL	Unmatched Conceptual Simplicity	Code Reusability & Semantic Richness
Ideal Use Case	Highly structured, standardized business intelligence.	Planning, communication, and visual translation.	Highly complex, interconnected, behavior-driven software systems.

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